Problem 6 – GPT-6 Astra

Abstract

Every Dehn surgery on an L-space knot in S^3 has at most two nontrivial prime summands. We prove this using established results about reducing slopes and prime decompositions, together with a group-theoretic genus bound for integer homology sphere summands of surgeries on fibered knots.

1 Statement and conventions

We use the standard ambient-manifold convention: the knots under consideration lie in S^3. A rational homology sphere Y is an L-space if

$$\dim_{\mathbb{F}_2} \widehat{HF}(Y;\mathbb{F}_2) = |H_1(Y;\mathbb{Z})|.$$

An L-space knot is a knot admitting a nonmeridional L-space surgery. We allow either sign for this surgery coefficient; when the coefficient can be chosen positive, we explicitly say that the knot admits a positive L-space surgery.

For a knot K, write $S^3_{p/q}(K)$ for Dehn filling along $p\mu + q\lambda$, where μ is a meridian and λ is the Seifert longitude. For finite slopes we take $\gcd(p,q) = 1$ and $q > 0$; the meridional slope is denoted by ∞.

All homology groups without specified coefficients have coefficients in $\mathbb{Z}$. Prime summands mean the non-S^3 factors in the Kneser–Milnor prime decomposition [Mil62, Theorem 1]. In particular, S^3 contributes no nontrivial prime summands.

Theorem 1. *Let $K \subset S^3$ be an L-space knot. For every $r \in \mathbb{Q} \cup \{\infty\}$, the manifold $S^3_r(K)$ has at most two nontrivial prime summands.*

We first state precisely the published surgery results that will be used, and then prove the required group-theoretic estimate.

2 Inputs from surgery theory

Theorem 2 (Fiberedness)**.** *If $K \subset S^3$ admits a positive L-space surgery, then its exterior fibers over S^1, with fiber a Seifert surface of genus $g(K)$.*

This is the fiberedness assertion of [BS22, Theorem 1], which supplies a proof of the result.

Theorem 3 (Hom–Lidman–Zufelt)**.** *Suppose that a nontrivial knot $K \subset S^3$ admits a positive L-space surgery and is not a cable knot, where nontrivial torus knots are counted as cables of the unknot. If $S^3_r(K)$ is reducible, then*

$$r = 2g(K) - 1.$$

We use the non-cabled formulation stated in the abstract of [HLZ15]. Their Theorem 1.3 establishes the hyperbolic case, and their Section 1 explains the reduction from the non-cabled case using the established torus- and satellite-knot results.

Theorem 4 (Howie)**.** *If a manifold obtained by Dehn surgery on a knot in S^3 has more than two nontrivial prime summands, then it has exactly three. Its prime decomposition has the form*

$$L_1 \# L_2 \# Z,$$

where L_1, L_2 are nontrivial lens spaces and $Z \ncong S^3$ is an integer homology sphere.

This is [How02, Corollary 5.3], including the consequences of the results of Valdez-Sánchez and Sayari incorporated in that corollary.

We will also use the Poincaré theorem: a closed simply connected 3-manifold is homeomorphic to S^3; see [MT07]. Consequently, a closed orientable 3-manifold with cyclic fundamental group has at most one nontrivial prime summand. Indeed, a decomposition into two non-S^3 factors would, by Seifert–van Kampen and the Poincaré theorem, express its fundamental group as a free product of two nontrivial groups. Such a free product is not cyclic: nonidentity elements from its two factors do not commute, by the reduced-word normal form.

Lemma 5 (The cable case)**.** *Every Dehn surgery on a nontrivial cable knot in S^3, including a nontrivial torus knot, has at most two nontrivial prime summands.*

Proof. It suffices to consider a nonprime surgery. The classical results for cable knots imply that such a surgery is along a cabling slope. For a nontrivial companion, this follows from [Sch90, Corollary 4.4 and the proof of Corollary 4.5]; the torus-knot case is contained in Moser's classification [Mos71].

Thus, for some knot J, we may write

$$K = C_{a,b}(J), \qquad r = ab,$$

where $a \geq 2$, $\gcd(a, b) = 1$, and a is the winding number of the cable. The cable-surgery formula [Gor83, Corollary 7.3] gives

$$S^3_{ab}(C_{a,b}(J)) \cong L \# S^3_{b/a}(J),$$

where L is a lens space with fundamental group of order a. In particular, L is prime by the preceding observation.

Since $a \geq 2$ and $\gcd(a, b) = 1$, the slope b/a is nonintegral. If J is nontrivial, the nonintegral-surgery theorem of Gordon–Luecke [GL87] says that $S^3_{b/a}(J)$ is irreducible, hence prime. If J is the unknot, its exterior has cyclic fundamental group, and the fundamental group of every filling is a quotient of that group. Thus $S^3_{b/a}(J)$ again has at most one nontrivial prime summand. The displayed connected sum therefore has at most two. □

3 A genus bound for homology sphere summands

For a finitely generated group G, let $\mathrm{rk}(G)$ denote the minimum size of a generating set. This is group-theoretic generating rank, not the rank of the abelianization.

Lemma 6. *Let $K \subset S^3$ be a fibered knot of genus g, and let p/q be a reduced finite slope with $p \neq 0$. Suppose*

$$S^3_{p/q}(K) \cong X \# Z,$$

where $Z \ncong S^3$ is an integer homology sphere. Then

$$|p| \leq g.$$

Proof. Set

$$E = S^3 \setminus \operatorname{int} \nu(K), \qquad Y = S^3_{p/q}(K), \qquad G = \pi_1(E), \qquad \Gamma = \pi_1(Y), \qquad n = |p|.$$

For a group A, write $A' = [A, A]$.

First, the commutator subgroup of Γ *has at most* $2g$ *generators.* The fibration of E has fiber a once-punctured genus-g surface F. Its homotopy exact sequence gives

$$1 \longrightarrow \pi_1(F) \longrightarrow G \longrightarrow \mathbb{Z} \longrightarrow 1.$$

The homomorphism to $\mathbb{Z}$ is the abelianization: a meridian maps to a generator, and $H_1(E) \cong \mathbb{Z}$. Hence

$$G' \cong \pi_1(F) \cong F_{2g},$$

where F_{2g} is the free group of rank $2g$.

Seifert–van Kampen identifies the filling homomorphism as the surjection

$$G \twoheadrightarrow G/\langle\langle \mu^p \lambda^q \rangle\rangle \cong \Gamma.$$

A surjective homomorphism maps the commutator subgroup onto the commutator subgroup: every commutator in the target is the image of a commutator of lifts. Therefore $G' \twoheadrightarrow \Gamma'$, and

$$\operatorname{rk}(\Gamma') \leq 2g. \tag{1}$$

Next, the homology sphere summand forces at least $2n$ *generators.* The standard surgery homology calculation gives

$$H_1(Y) \cong \mathbb{Z}/n\mathbb{Z}. \tag{2}$$

Indeed, $H_1(E)$ is generated by the meridian, the longitude is null-homologous, and filling imposes the relation $p[\mu] = 0$.

Because $Y = X \# Z$ and $H_1(Z) = 0$, the connected-sum homology formula gives

$$H_1(X) \cong \mathbb{Z}/n\mathbb{Z}.$$

Let $\widetilde{X} \to X$ be the connected n-sheeted covering corresponding to

$$\ker\bigl(\pi_1(X) \longrightarrow H_1(X)\bigr).$$

For $n = 1$, this is simply the identity covering.

Choose the balls used to form $X \# Z$, and write

$$X^\circ = X \setminus \operatorname{int} B^3, \qquad Z^\circ = Z \setminus \operatorname{int} B^3.$$

The inverse image of the ball in X consists of n disjoint balls in $\widetilde{X}$. Attach one copy of Z° to each of the resulting boundary spheres. This constructs a connected n-sheeted covering

$$\widetilde{Y} \longrightarrow Y$$

with

$$\widetilde{Y} \cong \widetilde{X} \# \underbrace{Z \# \cdots \# Z}_{n \text{ copies}}. \tag{3}$$

To identify the covering subgroup, observe that

$$\pi_1(Y) \cong \pi_1(X) * \pi_1(Z).$$

The covering monodromy is abelianization on the first factor and is trivial on the second, since $H_1(Z) = 0$. It is therefore precisely the abelianization

$$\Gamma \longrightarrow H_1(Y) \cong \mathbb{Z}/n\mathbb{Z}.$$

Consequently,

$$\pi_1(\widetilde{Y}) \cong \Gamma'.$$

The group $B = \pi_1(Z)$ is perfect because $H_1(Z) = 0$. It is nontrivial by the Poincaré theorem and the assumption $Z \ncong S^3$. A nontrivial perfect group cannot be generated by one element, since a one-generator group is abelian. Thus

$$\operatorname{rk}(B) \geq 2.$$

Applying Seifert–van Kampen to (3) gives

$$\Gamma' \cong \pi_1(\widetilde{X}) * \underbrace{B * \cdots * B}_{n \text{ factors}}.$$

Grushko's rank theorem states that generating rank is additive under free products of finitely generated groups; see [Sta65]. It follows that

$$\operatorname{rk}(\Gamma') = \operatorname{rk}\big(\pi_1(\widetilde{X})\big) + n \operatorname{rk}(B) \geq 2n. \tag{4}$$

Combining (1) and (4) yields $2n \leq 2g$, and hence $n = |p| \leq g$. □

4 Proof of the main theorem

Proof of Theorem 1. Mirroring changes surgery coefficients by a sign and reverses the orientation of the filled manifold:

$$S^3_{-r}(\overline{K}) \cong -S^3_r(K).$$

The L-space condition is invariant under orientation reversal, as is the number of prime summands. We may therefore assume that K admits a positive L-space surgery; compare [HLZ15, Section 1].

Meridional surgery gives S^3, so assume r is finite. If K is the unknot, its exterior has cyclic fundamental group, and every filling has cyclic fundamental group. The observation following Theorem 4 settles this case. If K is a nontrivial cable, Lemma 5 settles the claim.

We may consequently assume that K is nontrivial and non-cabled. Suppose, for a contradiction, that

$$Y = S^3_r(K)$$

has more than two nontrivial prime summands. In particular, Y is reducible. Put $g = g(K)$. By Theorem 3,

$$r = n = 2g - 1.$$

Since K is nontrivial, $g \geq 1$, so $n > 0$.

Theorem 4 supplies a decomposition

$$Y \cong L_1 \# L_2 \# Z,$$

where L_1, L_2 are nontrivial lens spaces and $Z \ncong S^3$ is an integer homology sphere. Let

$$a_i = |H_1(L_i)|, \qquad i = 1, 2.$$

Both a_i exceed 1. The surgery homology calculation and the connected-sum homology formula imply

$$n = |H_1(Y)| = a_1 a_2 > 1.$$

Together with $n = 2g - 1$, this shows that $g \geq 2$.

By Theorem 2, K is fibered. Apply Lemma 6 to the integer homology sphere summand Z, taking $X = L_1 \# L_2$. It gives

$$2g - 1 = n \leq g.$$

This is impossible for $g \geq 2$. The supposed decomposition cannot exist, proving the theorem. □

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